Towards Data-Driven Decision Making: The Convergence of Data Governance, Data Quality and Data Literacy

Abstract: *Data-driven approach is crucial in decision-making processes. Data becomes an asset and as a key-sight to credibility and trustworthiness. The convergence of Data Governance, Data Quality and Data Literacy in an organisation is the key element of success to establish a data-driven company that execute strategic decision based on analysis and interpretation of data. The need of Data Governance in recent data-driven era in a company to improve the database management process. Thus, the execution of data governance structure needs to be developed and assign the core data team to make sure the data gained is manageable and understandable across functional team, high quality and kept secured. Moreover, data quality is another key-success factor to a data-driven company where the data gained and used is 'fit for purposes' and produces excellent insight. High quality of data leads to trust of stakeholders to make good decisions while low quality of data leads to poor outcomes as per phrase 'Garbage In, Garbage Out'. Hence, it is important to become a data literate organization where the data are understandable, usage of analytical tools and compliance of data policy mindset.*

Keywords: *Data Governance, Data Governance Structure, Data Quality, Data Quality Dimensions, Data Literacy*

I. Introduction

In recent development of technology and business opportunities, data plays important roles in helping the business stakeholders in the state of decision-making to standout their company standards and profits. In order to achieve that, an organization must establish a data-driven company which strategic decision need to be made based on the analysis and the interpretation of data. There were several elements need to be implemented to achieve a data-driven organization such as data governance, data quality and data literacy. This paper will discuss and analyses on the significance of data governance in an organization, the roles and responsibilities of key people in data governance structure, high quality data leads to trustworthiness, characteristics of data quality that fit for purposes and accomplishment of data literate organization.

II. Data Governance in Demand

Data in an organization can be lots and unmanageable. Due to these problems, the needs of Data Governance implementation need to be executed as soon as possible to avoid future problems and low quality of data. Data Governance means the data needs to have a proper framework, guidelines, defined roles and comply the data policy standards of an organization. The author Friedman suggests that a holistic approach in an organization which focuses on "people, processes, and technology" need to be adopted to mitigate data quality issue. Hence, to ensure the quality of data is maintained, data needs to be governed (Friedman, 2006). This statement supports that data governance approach are in demand where many issues can be solved if every organization applied the data governance model.

Based on previous authors mentioned that the element of "people, processes and technology" needs to be adopted in managing the data, however it cannot be achieved if there is unclear defined roles and job scopes with maintaining data quality approach. As mentioned by Cheong & Chang, a high prestige company that execute an ad-hoc Data Governance approach found that data management efforts were obstructed due to lack of clarity of roles and responsibilities and lack of data quality mindset approach (Cheong & Chang, 2007). It declares that a systematic hierarchical structure with clearly defined roles needs to be established.

III. Key People in Data Governance Structure

Data Governance model without a proper structure that have clearly defined roles on each hierarchy of an organisation, will lead to confusion and goal by maintaining a good quality of data are impossible to achieve. The history of Data Governance Structure is demonstrated.

Informal data governance is formed namely 'Governance via Stewardship' (Thomas, 2006). But several reasons arise results the form of stewardship is not functioning for the Data Management team such as lack of power from upper executives such as the senior executives did not understand the importance of the project. Then, there is unclear roles and scope of works where there is no clarification of data stewards and data owners.

Consequences of that, there is lack of integration and connection between operational teams that acknowledge the problem and the teams who have power to make decision but are removed from the problem. Then, 'Governance via Governance' model was established to address the issue of decision-making authority.

The following figure shows that the Data Governance structure which comprise strategic level and operational based on roles and responsibilities.

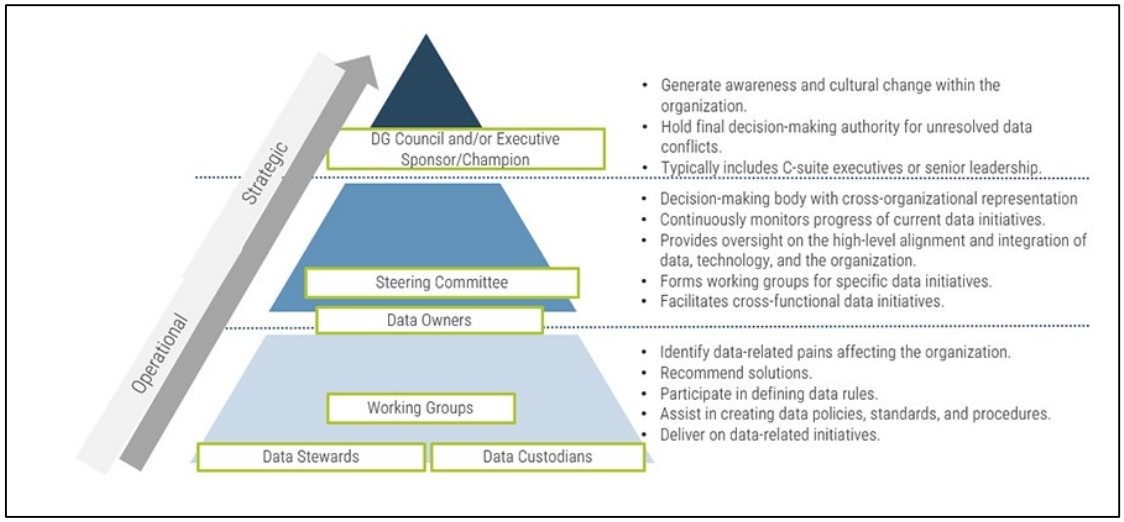

Figure 1 Traditional Data Governance Structure (https://www.infotech.com/research/ss/establish-data-governance)

Below the summary of roles and responsibilities mentioned by the author Chang & Cheong (2007).

Roles	Responsibilities
Data Governance Council	1. Authorize data policy making. 2. Solve issues among different functional departments. 3. Consult Information Technology (IT) council at strategic hierarchy. 4. Align business and IT initiatives. 5. Give feedback on financial budget submission for IT and non-IT related projects.
Data Custodian	1. Manage and ensure quality of data assets. 2. Solve issues raised in user group meetings 3. Authorize a plan of data management, data cleansing, ensure fit for purpose, change management and stakeholder management.
Data Steward	1. Knowledge of the process of business and requirements of data. 2. Understand IT knowledge and turn to business requirements. 3. Guide by Data Custodians to execute technical plans.
User Groups	1. Report any data related problems and issues. 2. Request functionality that would help them collect data more efficiently and specify reporting requirements.

IV. Trust in Data Begins with Data Quality

High quality of data guarantees the trustworthiness of data hence increase the confidence-level of decision maker. Cheong & Chang emphasized that a cornerstone for a successful data governance program is highly influenced by data quality. For example, the success of Control Objectives for Information and Related Technology (COBIT) framework which is used in publicly traded companies relies heavily on the quality of business data to ensure an accurate and credible financial report. While good data quality resulting in good decisions and reporting, low data quality would lead to poor outcomes as per phrase 'garbage in, garbage out'. For instance, poor data quality may increase the cost and complexity in the organization's efforts to streamline the business operations using software solutions such as customer relationship management (CRM) and supply chain management (SCM) (Cheong & Chang, 2007).

Cheong & Chang highlighted that the issues of poor data quality are mainly because of data are stored everywhere in various databases within an organisation (Cheong & Chang, 2007). Hence, it is recommended to adopt the concept of 'Single Source of Truth' which is a centralized data repository to guarantees the consistencies and reliability of data across the systems (Queiroz & Coltman, 2024)

Meanwhile, from the lens of data librarian, Koltay illustrated the roles of libraries to maintain data quality in a form of services. With the evolving role of libraries especially with the added feature of research data services, libraries could act as a 'data quality hubs' by offering auditing and verification services to ensure the authenticity, integrity and accuracy of the data over time (Koltay, 2016). Documenting metadata is also important to ensure sustainability and quality of research data as exact data origin and information about the data is recorded hence providing discoverability and facilitating the data integration process (Hassenstein & Vanella, 2022).

V. Data Quality Fit for Purposes

Data is considered as high quality if it 'fit for purpose' in which the data is able to fulfil the needs and requirements of data users to utilize the data. An organization needs to determine relevant characteristics to measure the quality of data which are the data quality dimensions. Up until now, there is not a universal agreed-to set of data quality dimensions, hence an organization is free to establish their own set of data characteristics that are worth to measure (DAMA International, 2017).

Koltay outlined some data quality dimensions which are relevant for academic librarian such as understandability and usability, completeness, applicability, acceptability, accuracy, availability and discoverability, trust and authenticity, and integrity (Koltay, 2017). In the context of business sector, Cheong & Chang pointed out six dimensions of data quality which are accuracy, completeness, timeliness, relevance, contextual definition, and trustworthiness (Cheong & Chang, 2007).

Based on a white paper produced by DAMA UK in 2013, it provided a selection of six important dimensions in data quality as per Figure 1 below.

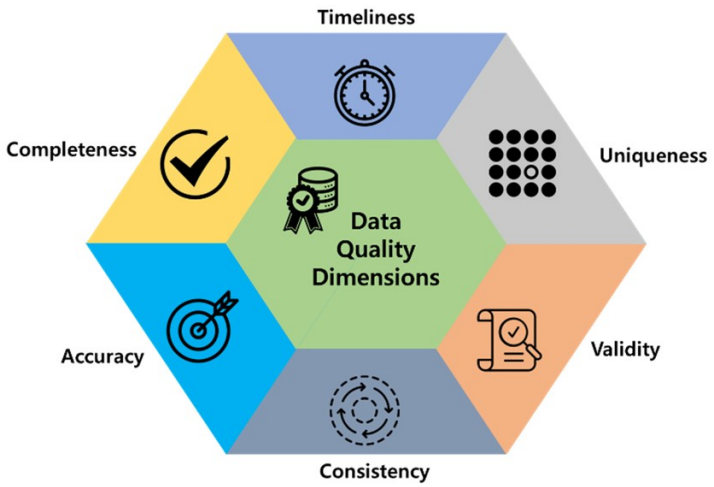

Figure 2: Data Quality Dimensions (Bhandari et al., 2021)

Here are the brief descriptions for each of the dimensions (DAMA International, 2017):

- Completeness: Measures whether all the required data are available.
- Uniqueness: Measures whether all the elements recorded are unique and no duplication.
- Timeliness: Measures whether the data is up to date to represent the reality.
- Validity: Measures whether the data is following the format or syntax set by the organization.
- Accuracy: Measures whether the data matches the real-world scenarios and confirms to verifiable sources
- Consistency: Measures whether the data is consistent when comparing two or more sources.

VI. Becoming a Data Literate Organization

In a simple word, data literacy can be defined as the ability to read, understand, produce, translate, and communicate data effectively. Koltay stated that it is important to upskill all employees with the ability to utilize and interpret data accurately to enhance decision making process that align with organizational goals. This is because, the lack of data literacy skills could lead to incorrect interpretation of data which resulting in misguided decisions. A notable example of data literacy issue can be seen from the case of the Mars Climate Orbiter in which a huge calculation error happened due to the wrong used of metric units (pounds) instead of NASA's standardized units (newtons) thus resulting in the failure to reach the desired altitude and major loss of $328 million (Koltay, 2017). Becoming a data literate organization will take some time and extra efforts, thus higher management should not treat it as an inconvenience; *it is not an inconvenience if it done correctly* (Ladley, 2020).

To embrace the culture of data literacy, Cheong & Chang stressed the needs for an organization to implement training and awareness for all data stakeholders to foster a mindset that values the quality of data with appropriate attention to data governance principles for a data-driven decision making. Even though Cheong & Chang did not clearly mention about data literacy, there are several contexts that can be indirectly related to data literacy. For example, in reference to the Data Governance Structure for Company A, it is apparent that data literacy skill is crucial in the cross functional collaboration between IT governance council and data governance council. This is because, both teams need to have a mutual understanding on shared data concepts to create a common language for communicating insights and strategies related to data governance initiatives (Cheong & Chang, 2017).

VII. Conclusion

In a nutshell, the convergence of data governance, data quality, and data literacy shape a comprehensive data governance framework that is crucial for the success of an organization in a data-driven era. Effective data governance structure delineates clear roles, responsibilities, and accountability so that data is not only well-governed but also reliable, high quality, and trustworthy. Data quality which measured by its dimensions such as timeliness, accuracy, and validity serves as a cornerstone of a successful data governance which is essential for deriving meaningful insights, enabling informed decision making, and crafting strategic initiatives. By adapting the culture of data literacy, the true value of data is fully realized once the employees

acquire the skills to interpret and utilize data efficiently, bridging the gap between data governance framework and practical compliances. This is making the integration between data governance, data quality, and data literacy vital for managing data as an asset.

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